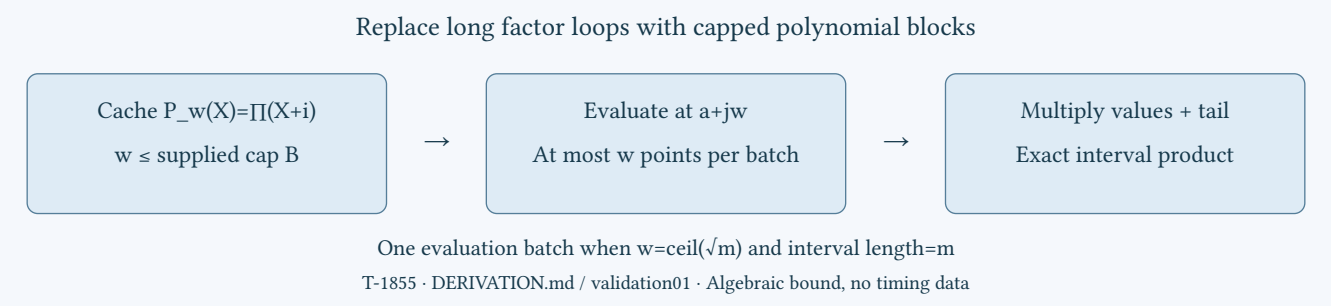
# Polynomial blocks for recurrence products

T-1855 · 2026-11-07 · Capped scalar-field interval arithmetic

The coefficient stream now evaluates long numerator/denominator products with a short cached polynomial and bounded multipoint batches. Short intervals retain direct multiplication. Symmetric binomial initialization also avoids long work for constant modes. Public-point source and recovery control flow are unchanged.




**Identity.** For R(a,l)=∏(a+i), split l=t w+u. Each full block is P_w(a+jw), with P_w(X)=∏(X+i), i=0..w−1. Their product times the final u factors equals R exactly, including zero factors and modular wrap. Substituting these exact products preserves every signed-binomial ratio and the prior recovery proof. Interval values are never inverted by the product helper.

**Conditional cost.** Choose w=min(ceil(sqrt(m)),B). The cached polynomial costs O(M(w) log w). At most ceil(floor(l/w)/w) evaluation batches each cost O(M(w) log w), plus floor(l/w) value products and a tail shorter than w. If B admits square-root width and l=m, this becomes O(M(sqrt(m)) log m+sqrt(m)) per product including setup. With the native Karatsuba model this is O(m^0.79249 log m). This is a derived arithmetic bound; fixed small B does not have that scaling, and the threshold32 is not a measured crossover.

| Native exact check | Count | Outcome |
|---|---|---|
| Interval comparisons | 4,820 | Equal direct multiplication |
| Blocked helper calls | 1,808 | Width/batch bounds hold |
| Zero / wrapped products | 2,485 / 1,305 | Exact boundary behavior |
| Constant-mode initializers | 3 | No block polynomial built |
| Additional public panel, m=56 | 281 | 14,421 calls; zero unresolved |
| Inherited public panel | 617 | 5,929 calls; zero unresolved |

**Storage and scope.** One polynomial of w+1 coefficients persists per stream; each temporary tree has O(w log w) coefficients, with kernel workspace separate. Mode-object storage remains32bytes. First C0 remains dense and mode-order validation costs O(m). The product cap applies even when the retained image degree is small. No end-to-end or observed timing claim follows from the per-product bound.

**Evidence.** validation01 pins46transitive sources and stdout; overflow checks enabled. Coefficient, residue and42,324state controls remain passing. PCC-20261107-8055fc is unsigned producer evidence. No native/report failures, cloud trial or full Cargo rerun. Original4,563-case cohort and prospective trace protocol unchanged.

**Sources and next construction.** Bostan-Gaudry-Schost2004 recurrence and multipoint-evaluation discussion, plus exact predecessor/native polynomial engine; KB unavailable. The implementation uses ordinary product/remainder trees, not the paper's specialized progression-shift algorithm. Next reduce repeated evaluation-tree construction or share evaluations across adjacent ratios. DERIVATION.md supplies proof, limits and replay; prior graphs remain unchanged.